
A Usability Evaluation Method for e-learning: Focus on Motivation to Learn

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Abstract

In this paper the development of a questionnaire-based usability evaluation method for e-learning applications is described. The method aims at extending the current practice by proposing intrinsic motivation to learn as a new usability measure to evaluate e-learning designs. The method was developed according to an established methodology in HCI research and relied upon a conceptual framework that combines web and instructional design parameters and associates them with a main affective learning dimension, intrinsic motivation to learn. Two large empirical studies were conducted in order to test the method. Results provide significant evidence for reliability of the method. Further work focuses on validation process so that usability practitioners can use it with confidence when evaluating the design of e-learning applications.

Keywords

Usability evaluation method, e-learning, users' affect, motivation to learn.

ACM Classification Keywords

H5.m. Information interfaces and presentation
H.5.2 User Interfaces - evaluation/methodology

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Introduction

The need to develop a usability evaluation method for e-learning applications

Evaluating the usability of e-learning applications is not a trivial task. Increase in the diversity of learners, technological advancements and radical changes in learning tasks, present significant challenges and render the possibility of defining the context of use of e-learning applications. Identifying who are the users and what are the tasks in e-learning context impose extra difficulties. As Notess [11] argues “*evaluating e-learning may move usability practitioners outside their comfort zone*”. Squires [15] highlights the need for integration of usability and learning [17] and points out the non-collaboration of workers in HCI and educational computing areas. In fact usability of e-learning designs is directly related to their pedagogical value; usability practitioners need to familiarize themselves with the educational testing research, learning styles and the rudiments of learning theory. Nevertheless very little has been done to critically examine the usability of e-learning applications; there is an ellipsis of research validated usability evaluation methods that address the user as a learner and consider cognitive and affective learning factors that facilitate the learners to achieve learning goals and objectives.

The emergence of new usability measures: the affective dimension

Traditional usability measures of effectiveness, efficiency and satisfaction are not adequate for new contexts of use such as technology supporting learning [4]. It is critical that systems designers assess the range of possible affective states that users may experience while interacting with a software application. To this end new usability techniques and

measures need to be established. In the e-learning context affect has recently gained attention. It has been argued that affect is the fuel that learners bring to the learning environment connecting them to the “*why*” of learning. New developments in learning theories stress the need to enhance learners’ internal priorities and drives that can be best described by motivation to learn [13]. The latter, a concept intimately linked with learning [13, 14], is the most prominent affective learning factor that can greatly influence users’ interaction with an e-learning application. Intrinsic motivation to learn is proposed as the anchor for the development of a new usability measure for e-learning designs. Thus, the purpose of this research is to develop and empirically test a questionnaire-based usability evaluation method for e-learning applications. This method addresses the user as a learner and integrates both cognitive (users’ perception of usability) and affective aspects (users’ motivation to learn). The development of the method was based upon a very well known methodology in HCI research and practice [8]. Two large empirical studies were conducted examining usability of e-learning applications in authentic environments, with real learners-employees in corporate settings. The focus was on empirical assessment of the reliability of the method. Next section describes the development of the method followed by discussion on further work.

Method Development

The Questionnaire Design Methodology suggested by Kirakowski and Corbett [8] was followed throughout the development of the proposed method. This methodology contains five stages: 1) Forming the survey, 2) Item sampling, 3) Pilot trial, 4) Production version and 5) Next version. This paper reports the

work conducted along the first three stages, while the other stages is an issue of work-in progress.

Forming the survey

Regarding the “Forming the survey” stage, two objectives need to be considered: a) the type, and b) the scope of the questionnaire. A psychometric-type of questionnaire was considered as the most suitable method since the major objective was to measure users’ perception of e-learning usability and related affect (in this case motivation to learn). As far as concerns the scope of the questionnaire, it has to be noted that this research focuses on usability evaluation of asynchronous e-learning applications as adult learners use them in corporate settings for training purposes. In addition this method extends the current usability practice by focusing on motivation to learn as an important dimension so as to address the user as a learner. This research relies mainly on Keller’s [7] theory and related model, which is perhaps the most influential model of motivation, in order to further analyze and interpret the motivation to learn construct. This construct is composed of four sub-constructs: attention, relevance, confidence and satisfaction. Prior to the development of the questionnaire a conceptual framework was established [16], which employs a) a combination of web design and instructional design parameters and b) motivation to learn construct. According to the conceptual framework usability parameters that were chosen from an array of studies for inclusion into the questionnaire are: Navigation, Learnability, Accessibility, Consistency, Visual Design, Interactivity, Content & Resources, Media Use, Learning Strategies Design, Instructional Feedback, Instructional Assessment and Learner Guidance & Support. Thus a new set of usability parameters (derived from web

usability and instructional design literature) for e-learning is proposed along with a new type of usability dimension: motivation to learn.

Item sampling

The design parameters included in the conceptual framework were the main constructs included in the questionnaire. These constructs were measured with items adapted from prior research. To identify items for possible inclusion in the questionnaire, an extensive review of prior studies referring to e-learning and web design was conducted. More specifically a number of web and instructional design guidelines [5, 8, 9, 10] have been reviewed, as well as a number of usability evaluation heuristics, checklists and questionnaires [5, 12]. Items were carefully selected so that to cover all parameters included in the conceptual framework. The questionnaire development started with an initial item pool of over 90 items. The items were examined for consistency of perceived meaning by getting 3 subject matter experts to allocate each item to content areas. Some items were eliminated when they produced inconsistent allocations. Prior to completion of the first version of questionnaire, a mini pilot test was undertaken to ensure that items were adapted and included appropriately in the questionnaire. An online version of the questionnaire was administered to 15 users in academic settings (2 of them were usability experts) that had some prior experience with e-learning courses. Data obtained was analyzed mainly for response completeness; some adjustments were made and subsequently some items were reworded. The whole procedure led to the first version of questionnaire, which consisted of 64 items: 54 items measuring web and instructional design usability and 10 items measuring motivation to learn. Criteria

corresponding to each usability parameter were assessed on a 5 point scale, where the anchors were 1 for strongly disagree and 5 for strongly agree. There was also space for free-form comments. The first version of questionnaire was tested in pilot trial 1.

Pilot trial 1

The first version of the questionnaire was used and empirically tested during an international research project. Four user-organizations participated in the project and used a common e-learning service in order to train a subset of their employees (mostly IT professionals) in the following topics: "IT Business Consultancy" and "Software and Applications Development". A summative type of usability evaluation was conducted after the implementation of the e-learning service. The trainees interacted with the e-learning service during a period of four months and then they were asked to participate in the summative evaluation study. A web-based version of the questionnaire was developed which was released right after the end of the project's formal training period. The respondents self-administered the questionnaire and for each question, were asked to circle the response which best described their level of agreement with the statements. The total number of trainees who participated in usability evaluation was one 113 (63 male and 50 female).

ANALYSIS AND RESULTS

First a factor analysis was conducted in order to identify the underlying dimensions of usability parameters of e-learning courses as perceived by the trainees. As a result an 8-factor solution was extracted explaining 65,865% of the variance. In order to assess the reliability of the instrument Cronbach's Alpha was

utilized. The findings reveal that all factors show high internal consistency as indicated by high Alpha coefficients (ranges from 0.707 to 0.901), which exceed the recommended level of .70 [3]. In addition the composite variable *Motivation to Learn* shows a very high internal consistency as Alpha coefficient indicates ($\alpha = 0.926$). After conducting factor and reliability analysis a new version of the questionnaire (version 2) was derived. Data analyses led to the refinement of the questionnaire and a more parsimonious solution has been reached with 8 factors representing usability parameters of e-learning courses: Interactive content, Instructional Feedback & Assessment, Navigation, Visual Design, Learner Guidance & Support, Learning Strategies Design, Accessibility and Learnability. The second version of the questionnaire contains 41 items representing 8 usability parameters (the 8 factors extracted as already presented) and 10 items representing the *Motivation to Learn* construct, a total of 51 items. This new version of the questionnaire, which shows a very high reliability as measured by Cronbach Alpha coefficient (0.961) is used and empirically tested in the second large-scale study (pilot trial 2) to be described in the next section.

Pilot trial 2

The second large-scale study was conducted in collaboration with a large private Greek bank, which had implemented e-learning services and courses as a supplementary to traditional mode of training. The second version of the questionnaire was paper-based and it was distributed in Greek language. The target population of the study was the whole set of employees that were trained via e-learning courses. The respondents self-administered the questionnaire and for each question, were asked to circle the response

which best described their level of agreement with the statements. Data from two 256 respondents were actually used in the analysis. The trainees who responded were bank tellers and financial planners. Among them 110 were male and 146 were female.

ANALYSIS AND RESULTS

Forty-one items representing eight usability parameters were factor analyzed using principal components method with a Varimax rotation. After performing two iterations of factor analysis, the pattern of factor loadings suggested that a seven-factor solution should be extracted. The seven-factor solution explained 54% of the variance. These factors were: Content, Learning & Support, Visual Design, Navigation, Accessibility, Interactivity and Self Assessment & Learnability. For the assessment of reliability Cronbach's Alpha was utilized. All factors showed high internal consistency as indicated by high Alpha coefficients, which exceed the recommended level of 0.70 [3]. In addition the composite variable *Motivation to Learn* shows a very high internal consistency as Alpha coefficient indicates ($\alpha=0.900$). After conducting factor and reliability analysis a new more parsimonious solution emerged and a new version (version 3) of the questionnaire was derived. The overall alpha for the new version of questionnaire was very high, $\alpha=0.934$. This version of questionnaire contains 39 items measuring e-learning usability parameters plus 10 items measuring *Motivation to Learn*, a total of 49 items.

Further study

Motivated by the need to address the specificities of e-learning design a usability evaluation method was developed. The proposed questionnaire-based usability

evaluation method extends conventional web usability criteria and integrates them with criteria derived from instructional design so that to address specificities of e-learning design and address the users as learners. The proposed method also extends the current practice in usability evaluation by measuring users' affective engagement and proposing *motivation to learn* as a new type of usability measurement. This new type of measurement has been tested for reliability and overall internal consistency of the questionnaire is very high. The method can also be used by the e-learning designers and usability practitioners as a formative evaluation method for benchmarking purposes.

The development of the method is an ongoing process; current work is focused on the assessment of validity of the method. It is widely recognized that validation is a process [4] and adequate evidence for construct validity cannot be obtained from a single study but from a number of inter-related studies [1]. The method is going to be used in different studies with a number of e-learning courses, different types of learners, different corporate environments. Regarding intra-questionnaire relationships several statistical analyses can be used such as multiple regression analysis in order to justify the quantitative association between usability properties and motivation to learn or confirmatory factor analysis in order to determine convergent and divergent (or discriminant) validity [3]. Learning assessment will also be used as an external validity criterion. Besides the assessment of validity additional research effort will focus on the development of the next version of the questionnaire (stage 5); the goal is to seek a deeper understanding of the design issues that influence learners' affect and emotions. Emotions such as fear, anxiety, apprehension, enthusiasm and

excitement as well as pride and embarrassment along with the flow experience [2] can provide significant input to e-learning design and shed light in explaining learners' behaviour; such constructs are expected to enhance the power of the proposed method and will be taken into consideration in the next version of the questionnaire.

REFERENCES

- [1] Anastasi, A., and Urbina, S. *Psychological testing*. 7th Edition, Prentice Hall, 1997.
- [2] Csikszentmihaly, M. *Flow: The psychology of optimal experience*. New York: Harper & Row, 1990
- [3] Hair, J. F., Anderson, R. E., Tatham, R. L. T., and Black, W. C. *Multivariate data analysis*, Prentice-Hall Inc., Upper Saddle river, New Jersey, 1998.
- [4] Hornbaek, K. Current practice in measuring usability: Challenges to usability studies and research. *International Journal of Human-Computer Studies* (in press), 2005.
- [5] Horton, W. *Designing Web-Based Training*. John Wiley & Sons, 2000.
- [6] Johnson, S., and Aragon, S. An Instructional Strategy Framework for Online Learning Environments. In *Proc. of World Conference on E-Learning in Corp., Govt., Health., & Higher Ed*. Vol. 2002, (1), 529-536.
- [7] Keller, J.M. Motivational design of instruction. In Reigeluth, C. M. (ed.), *Instructional Design Theories and Models: An overview of their current status*. Hillsdale, NJ: Erlbaum, 1983.
- [8] Kirakowski, J. and Corbett, M. *Effective Methodology for the Study of HCI*. Elsevier, 1990.
- [9] Lynch, P.J., and Horton, S. Interface design for WWW Web Style Guide. Yale Style Manual. <http://info.med.yale.edu/caim/manual/interface.html>.
- [10] Nielsen, J. *Designing Web Usability: The Practice of Simplicity*. New Riders: New York, 2000.
- [11] Notess, M. Usability, User Experience, and Learner Experience. <http://www.elearnmag.org/>.
- [12] Reeves, T. C., Benson, L., Elliott, D., Grant, M., Holschuh, D., Kim, B., Kim, H., Lauber, E., and Loh, S. Usability and Instructional Design Heuristics for E Learning Evaluation. In *Proc. of World Conference on Educational Multimedia, Hypermedia and Telecommunications*. Vol. 2002, 1 (2002), 1615-1621.
- [13] Schunk, D. *Learning theories: An educational perspective* (3rd Ed). Upper Saddle river, NJ: Prentice-Hall., 2000.
- [14] Soloway, E., Guzdial, M., and Hay, K. Learner-Centered Design. The Challenge For HCI In The 21st Century. *Interactions*. 1, 2 (1994), 36-48.
- [15] Squires, D. Usability and Educational Software Design: Special Issue of Interacting with Computers. *Interacting with Computers*. 11, 5 (1999), 463-466.
- [16] Zaharias, P. E-Learning Design Quality: A Holistic conceptual framework. To appear in *Encyclopedia of Distance Learning*. Edited by Howard et al., Idea Group Inc., New York, 2005.
- [17] Zaharias, P. Usability and e-Learning: The road towards integration. *ACM eLearn Magazine*, Vol.2004 (6).